

Transit Light Curve Analysis of WASP-12b with TASTE and TESS observations

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March 29, 2026

ABSTRACT

This article presents the results of the transit light curve analysis of WASP-12b, utilising observational data acquired from the TESS telescope during Sectors 43, 44, and 45, as well as data from the TASTE campaign. We used Python notebooks with complex packages such as `batman` and `emcee`. This research validates the designation of WASP-12b as a Hot Jupiter, exhibiting a radius of $1.88 \pm 0.016 R_J$ and a transit depth of 0.013715 ± 0.000152 . The orbital period was refined to 1.091418 ± 0.000001 days. Disparities with published values were observed in parameters such as the semi-major axis ratio (a/R_s) and orbital inclination (i), likely attributable to variations in modelling assumptions or detrending techniques. The phase-folded TESS light curves displayed a minor misalignment, indicating possible enhancements in T_0 estimation or correction of residual systematics.

1. Introduction

Recent exoplanet discoveries and studies have improved our knowledge of planetary systems. Observational methods include gravitational microlensing, astrometry, transit photometry, radial velocity, and direct imaging have helped find exoplanets. The transit photometry approach has successfully identified and characterised several exoplanets (Haswell 2010; Seager 2011). This approach detects transits, which occur when a planet covers a star and reduces its light. These dimming events can be used to detect a planet by studying the star's light curve, which shows its brightness over time. This approach is useful because it measures basic planetary characteristics. Examples include the planet's orbital period, chemical makeup, and transit depth (Haswell 2010). The transit method can also reveal a planet's habitability. Integrating orbital and atmospheric data with stellar attributes helps us assess a planet's life-sustaining capacity by assessing surface temperatures and atmospheric conditions. According to (Seager & Deming 2010) and (Winn 2010), the transit method is a widely used approach in exoplanetary science, essential for comprehending distant planetary systems. This research investigates the exoplanet WASP-12b, a transiting short-period gas giant identified and characterised by the Wide Angle Search for Planets (WASP) mission in 2008 (Hebb et al. 2009). After finding transit candidates, more observations are needed to validate their planetary features. Astronomers can use radial velocity to discover a planet's gravitational pull on its host star. This eliminates false positives, such as eclipsing binaries, and confirms the planet's existence (Pont 2005). Spectroscopic study also yields precise information on the atmosphere of the host star. Effective temperature, surface gravity, metallicity, and projected rotational velocity are obtained by fitting observed spectra to artificial templates. This characterisation is necessary to comprehend the star and its planetary system (Biazzo et al. 2022).

WASP-12b is a hot Jupiter, a gas giant like Jupiter with a shorter orbital period and higher surface-atmosphere tem-

peratures owing to its closeness to its host star. The planet orbits an F-type host star in 1.091 days, making it one of the shortest-period hot Jupiters. WASP-12b, possessing a mass of around $1.41 M_J$ and a radius of $1.9 R_J$ according to NASA Exoplanetary Archive, exhibits a significantly inflated atmosphere, perhaps due to severe star irradiation. The semi-analytic model by Mandel & Agol (2002), utilising limb-darkening coefficients from Claret (2004), is commonly used to characterise the transit light curves based on the features of the host star. The orbital configuration corresponds to a circular path. Significant tidal interactions caused by WASP-12b's closeness to its host star cause atmospheric escape and cause the planet to become somewhat ellipsoidal (Li et al. 2010).

This work aims to reassess the radius and additional properties of WASP-12b utilising two distinct data sets, as detailed in Section 2. Asiago Search for Transit Timing Variation of Exoplanets (TASTE) initiative provided the initial dataset. TASTE, started in 2010, uses high-precision differential photometry at the Asiago 1.82m telescope to determine transit time variations. The other dataset came from NASA's Transiting Exoplanet Survey Satellite (TESS), which uses the transit technique to find exoplanets circling nearby stars. TESS detects short-period exoplanets and provides high-fidelity light curves for transit analysis. Section 3 describes how WASP-12 light curves from these two sources were processed and selected to locate transits. In Section 4, we concentrate on fitting these light curves by integrating limb darkening corrections using a quadratic stellar intensity model, together with jitters and polynomial factors. Section 5 delineates the findings of our investigation, followed by a discussion of their ramifications and a comparison with the published parameters from (Ozturk & Erdem 2019).

2. Data and Observations

2.1. TASTE

The raw observational data from TASTE including bias, flat-field and, science frames, were stored in FITS (Flexible Image Transport System) format. Important material about the observational setup, environmental conditions, and data processing parameters is found in the FITS header. From the header, we find the date of observation for our dataset to be 2019-12-29. The calibration of the raw data began with the acquisition of bias and flat-field frames during the observational session. Bias frames were obtained by capturing images with zero exposure time. These frames account for the instrumental signal introduced by the read-out current in the CCD detector. Multiple bias frames were averaged to produce a master bias frame, which was subsequently subtracted from all raw science images to eliminate this systematic offset. Flat-field frames were employed to compensate for non-uniformities in the CCD's reaction to light caused by variances in pixel sensitivity, dust on the optics, vignetting, or other optical path imperfections. By aiming the telescope at a long, uniformly lit source, such the twilight sky or lamps shining on a dome screen, flat-field frames were obtained. These flat-field frames were stacked and normalised to produce a master flat following bias correction. In order to normalise the scientific pictures and guarantee precise flux measurements in relation to pixel efficiencies, this master flat functioned as a pixel-wise correction factor. In Section 3.2, comprehensive methods for flat-field analysis are presented.

2.2. TESS

To support the analysis of the TASTE dataset, TESS data from adjacent sectors 43, 44, and 45 had to be used. These sectors' continuous coverage (see table 1), each with a 120-second exposure period, allowed for thorough temporal monitoring of the transits and lessened biases brought on by missing data. It has facilitated cross-validation of transit parameters, hence improving the consistency and dependability of the results. It also allowed us to enhanced the signal-to-noise ratio (SNR) and diminish uncertainty in critical parameters while alleviating sector-specific systematic errors. TESS offers various data products; however, we are just interested in the following products: Target Pixel File (TPF) and Light Curve File (LCF) which are accessible for each sector. This autonomous dataset validated the dependability of the TASTE project findings and enhanced the comprehensive analysis.

Table 1. Observation Sectors and Start Dates

Sector	Observation Start Date
43	2021-09-16
44	2021-10-12
45	2021-11-07

3. Methods

3.1. TASTE Dataset

3.1.1. Bias Correction

Bias is a constant value that is added to the system during data acquisition. This offset is non-astrophysical and must be removed to ensure accurate analysis. The primary source of uncertainty in the bias correction arises from read-out noise (RON), an inherent limitation of CCD performance. RON introduces perturbation whenever a bias frame is read. To quantify the impact of RON, 30 bias frames were collected. Each bias frame is a 2D image with dimensions 222×521 . After converting the data to electron units using the CCD gain factor (1.91 e/ADU), the bias frames were stacked to form a 3D array. The bias for each pixel was calculated by taking the median of all frames. The median approach was chosen over the mean because it is more robust against outliers, as evidenced by comparing the distributions of mean and median values for all pixels. We calculated associated error for the bias correction by calculating the standard deviation of each pixel in all bias frames. Then we divided the standard deviations by the square root of the number of frames ($\sqrt{N_{\text{frames}}}$). The final error, σ_{MB} , was determined using the equation:

$$\sigma_{\text{MB}} = \frac{\sigma_{\text{single}}}{\sqrt{N_{\text{frames}}}}, \quad (1)$$

where σ_{single} is the standard deviation of a single bias frame. The CCD's readout noise (RON) was measured at 7.10 e, and the standard deviation of a single bias frame in the region of interest (columns 300-350) was estimated as 7.49 e. This value is slightly higher than the RON, indicating the presence of additional variability in the bias frames. The median bias frame's estimated noise was determined using the formula:

$$\sigma_{\text{MB}} = \frac{\sigma_{\text{RON}}}{\sqrt{N_{\text{frames}}}}, \quad (2)$$

The value obtained was 1.30 e. The median bias frame's estimated standard deviation (calculated from the single-frame standard deviation) was 1.37 e. However, the measured standard deviation of the median bias frame was slightly larger at 1.69 e, possibly due to the usage of the median rather than the mean during stacking. To quantify uncertainties, we estimated the median pixel error throughout the whole master bias frame as 1.31 e. This consistency across individual pixels indicates the durability of the error propagation mechanism and ensures the accuracy of bias correction. These error estimates are crucial in downstream data reduction, providing a solid foundation for precise scientific frame calibration. By rigorously measuring the uncertainties associated with bias correction, this procedure ensures that the derived astrophysical data is free of systematic mistakes introduced during acquisition.

3.1.2. Flat-Field Correction

The flat-field correction procedure began with the extraction of data from the flat-field directory. We transformed the unprocessed pixel data into electron counts utilising the CCD gain factor. We subsequently, examined a solitary flat-field image by rendering the unprocessed pixel values

as an image. To enhance the visual representation of the data, we focused on the overscan zone, characterised by a markedly reduced count in the image's periphery. The overscan zones do not influence data reduction or scientific analysis. We calculated the lowest and maximum counts after omitting the overscan regions, and the outer columns were consistently eliminated from all further analyses. Next, we measured the normalization factors. We then, created a 3D array, referred to as stack, to store all flat-field images, with dimensions corresponding to the number of images and the dimensions of each frame. Specifically, the dataset consisted of 30 images, each with dimensions 521×222 pixels. Following that, we subtracted the median bias frame from each flat-field image to eliminate bias contributions. Since our primary focus is on the relative pixel behaviour rather than absolute efficiency, the flat correction was expressed as values relative to the median response of a selected pixel sample. To establish a reference value for each frame, we calculated the median intensity within a 50×50 pixel region at the centre of the 521×222 frame, resulting in a normalization factor. We then normalized each frame in the stack by dividing it by this factor. Finally, we computed the master flat-field as the median across all normalized frames in the stack. We computed associated error through error propagation. The primary sources of noise contributing to the flat-field error were: Readout Noise (σ_{RON}), The error associated with the median bias frame (σ_{MB}) and, Photon noise due to the lamp flux (σ_{PN}). The total error of the normalized flat-field was computed using:

$$\sigma_{\text{Normalized Flux}} = \frac{\sqrt{\sigma_{\text{RON}}^2 + \sigma_{\text{MB}}^2 + \sigma_{\text{PN}}^2}}{n}, \quad (3)$$

where n is the normalization factor. We calculated the photon noise for each frame as the square root of the flux ($\sigma_{\text{PN}} = \sqrt{\text{FLUX}}$). The error associated with the median master flat-field can be estimated by approximating the median function with the mean function. Although the median function does not have a closed-form analytical expression, the mean function does, making it suitable for use with the error propagation formula. The formula for that is the following:

$$\sigma_{\text{MF}} = \frac{\sqrt{\sum_{i=1}^{N_{\text{frames}}} \sigma_{\text{Normalized Flux}}^2}}{N_{\text{frames}}} \quad (4)$$

3.1.3. Science Frames Correction

The scientific images utilised in this investigation are unprocessed data acquired directly from the CCD detector. We transformed the raw data from (ADUs) to electrons by multiplying the value by the detector gain factor, as performed for bias and flat-field, so guaranteeing the data were in physically significant units of electrons. We subsequently adjusted the images for instrumental effects, encompassing bias and flat-field corrections. We accomplished the Bias Correction on the science frames by deducting the bias contribution from each frame, whereas we executed the flat-fielding by dividing all frames by the master flat-field to rectify pixel-to-pixel sensitivity discrepancies. Multiple error sources contribute to the comprehensive error budget of the corrected frames. This encompasses readout noise, errors linked to the bias frame, and photon noise. To quantify

these uncertainties, we initially calculated the error linked to the de-biased frames $\sigma_{S_{\text{deb}}}$ using the subsequent expression:

$$\sigma_{S_{\text{deb}}} = \sqrt{\sigma_{\text{RON}}^2 + \sigma_{\text{MB}}^2 + \sigma_{\text{PN}}^2} \quad (5)$$

Next, we calculated the error on the corrected scientific frames $\sigma_{S_{\text{deb,def}}}$. This accounts for additional uncertainties introduced during flat-field correction. The error is expressed as:

$$\sigma_{S_{\text{deb,def}}} = S_{\text{deb}} \sqrt{\left(\frac{\sigma_{S_{\text{deb}}}}{S_{\text{deb}}}\right)^2 + \left(\frac{\sigma_{\text{MF}}}{MF}\right)^2} \quad (6)$$

Here, S_{deb} is the signal after bias correction, $\sigma_{S_{\text{MF}}}$ is the uncertainty associated with the master flat-field, and MF is the master flat-field itself. This systematic approach ensures that all significant contributions to the error budget are accounted for, providing a robust basis for subsequent scientific analysis.

In astronomical observations, the recorded time of an event is expressed in Julian Date (JD), which represents the moment the exposure began at the observatory. However, for precise timing analysis, we need to correct for Earth's motion and convert the observation time to Barycentric Julian Date (BJD), referenced to the Solar System Barycenter (SSB). This correction eliminates the effects of Earth's movement in its orbit around the Sun. The conversion from JD to BJD TDB (Barycentric Dynamical Time at the Barycenter) involved computation of mid-exposure time, conversion from UTC to TDB and an application of the light travel correction. Each frame's Julian Date records the start of the exposure. To obtain the most accurate representation of the observation time, we computed the mid-exposure JD by adding half of the exposure duration:

$$JD_{\text{mid}} = JD_{\text{start}} + \frac{\text{Exposure Time}}{2 \times 86400} \quad (7)$$

where JD_{start} is the recorded Julian Date at the start of the exposure, Exposure Time is given in seconds, the denominator 86,400 converts seconds into fractional days. The next step is to shift from Coordinated Universal Time (UTC) to Barycentric Dynamical Time (TDB). This is necessary because UTC includes leap seconds, making it discontinuous, TDB is a relativistic coordinate time scale, continuously defined at the barycentre. The conversion was handled by `astropy`'s time module, ensuring the correct application of time scale transformations. Since Earth is in motion, photons travel different distances depending on Earth's position in its orbit. We correct for this by computing the light travel time from Earth to the Solar System Barycentre:

$$\Delta t_{\text{ltt}} = \frac{d_{\text{Earth-SSB}}}{c} \quad (8)$$

where $d_{\text{Earth-SSB}}$ is the instantaneous distance between Earth and the Solar System Barycenter (SSB), obtained from JPL ephemerides, c is the speed of light. The corrected time in BJD TDB is then:

$$BJD_{\text{TDB}} = JD_{\text{mid}} + \Delta t_{\text{ltt}} \quad (9)$$

The computed values ensure accurate barycentric time corrections for high-precision photometric and spectroscopic measurements. The average light travel time correction in this study was 8.19 minutes, with a mean difference between JD UTC and BJD TDB of -560.67 seconds. By performing this conversion, we obtain an accurate measurement of the event timing, free from distortions introduced by Earth's movement and leap second discontinuities in UTC. This is crucial for studies such as exoplanet transits.

3.1.4. Centroid Measurement

To achieve accurate photometric measurements, we calculated the centroids of the target and reference stars via an iterative refinement method. This approach improves the precision of star placement within the photometric aperture while also accounting for pixel sensitivity changes. The centroid coordinates, (x_c, y_c) , were initially determined using the flux-weighted mean formula:

$$x_c = \frac{\sum(x_i \cdot f_i)}{\sum f_i}, \quad y_c = \frac{\sum(y_i \cdot f_i)}{\sum f_i}, \quad (10)$$

Where x_i, y_i are the corresponding pixel coordinates and f_i is the flux recorded in pixel i . We carried out this calculation out within a circular aperture centred on the star's initial position and we iteratively modified the centroid to improve precision. During each iteration, the target distance from the current centroid is calculated for all pixels in the aperture. A new flux-weighted centroid is then produced using the revised pixel selection. The iterations continued until the percentage variation in both x_c and y_c between successive iterations was less than 0.1%. The target star's initial coordinates were $(x, y) = (357.00, 37.00)$. After three iterations, we identified the refined coordinates as $(x, y) = (357.00, 37.64)$, with the least deviation between iterations. This improvement greatly increased the precision of the star's centroid, minimising flux measurement errors and increasing the dependability of the generated light curves. In addition, we determined Full Width at Half Maximum (FWHM) along both axes of the star's profile to evaluate its shape and symmetry. For the target we obtained FWHM values of 7.94 pixels on the X-axis and 7.70 pixels on the Y-axis, showing a nearly symmetric light distribution. We subsequently reperformed all the previous steps for two reference stars', the refined coordinates for each are (172.21, 186.39) and (471.12, 90.58) respectively.

3.1.5. Aperture Photometry

We measured our target star's flux using aperture photometry on the sky-background corrected frames. The target star's improved centroid coordinates, $(x, y) = (357.00, 37.64)$, served as the aperture's centre. We computed the cumulative flux as a function of radius and normalised it to the total flux in the inner region in order to find the ideal aperture radius. To determine the radius that captures 95% of the total stellar flux, we created a plot of fractional flux vs aperture radius. This investigation led to the selection of

an aperture radius of 7 pixels as shown in figure 2. An annular region surrounding the target star with inner and outer radii of 13 and 20 pixels, respectively, this annulus included 722 pixels which we used to estimate the sky-background flux. After calculating the median and average fluxes in the annulus, the corresponding error was calculated as follows:

$$\sigma_{\text{sky}} = \sqrt{\frac{\sum \sigma_{\text{pixel}}^2}{N_{\text{pixels}}}}, \quad (11)$$

where N_{pixels} is the number of pixels in the annular region. Within the predicted uncertainty, the median and average sky fluxes for the target star were consistent. We then calculated the net stellar flux as:

$$F_{\text{net}} = F_{\text{aperture}} - (N_{\text{pixels}} \times F_{\text{sky}}), \quad (12)$$

where F_{aperture} is the total flux within the aperture, F_{sky} is the median sky-background flux per pixel, and N_{pixels} is the number of pixels in the aperture. The error on the flux was determined as:

$$\sigma_F = \sqrt{F_{\text{aperture}} + \sigma_{\text{sky}}^2 \times N_{\text{pixels}} + \sigma_{\text{readout}}^2}, \quad (13)$$

where σ_{readout} accounts for the CCD readout noise. By successfully separating the stellar flux and reducing contamination from the sky background and surrounding stars, this technology guaranteed accurate photometric readings. A strong basis for light curve analysis was given by the resultant fluxes and their uncertainty. Figure 1, illustrates the annuli centred on the refined locations of the target and the two reference stars.

The fluxes, corrected for instrumental effects and background contamination, along with their associated uncertainties, were the basis for our light curve study. This rigorous photometric extraction ensured high-quality data, appropriate for the modelling and analysis of the planetary transit.

3.1.6. Light Curve Extraction

We subsequently derived differential light curves to account for fluctuations in atmospheric transparency and other systematic influences by normalizing the flux of the target star using reference stars. The differential flux for each frame was calculated as follows:

$$DF = \frac{F_{\text{target}}}{\sum F_{\text{reference}}}, \quad (14)$$

where F_{target} refers to the net flux of the target star, and $\sum F_{\text{reference}}$ is the total flux of the chosen reference stars. This normalization helps mitigate variations from frame to frame, resulting in a more consistent light curve. To enhance the reliability of the differential flux measurement, we meticulously selected reference stars based on their brightness and stability, ensuring they demonstrated minimal variability throughout the observation period. These stars were also chosen to have brightness similar to that of the target star to reduce the effects of differential atmospheric extinction. The uncertainties in the differential flux were calculated using:

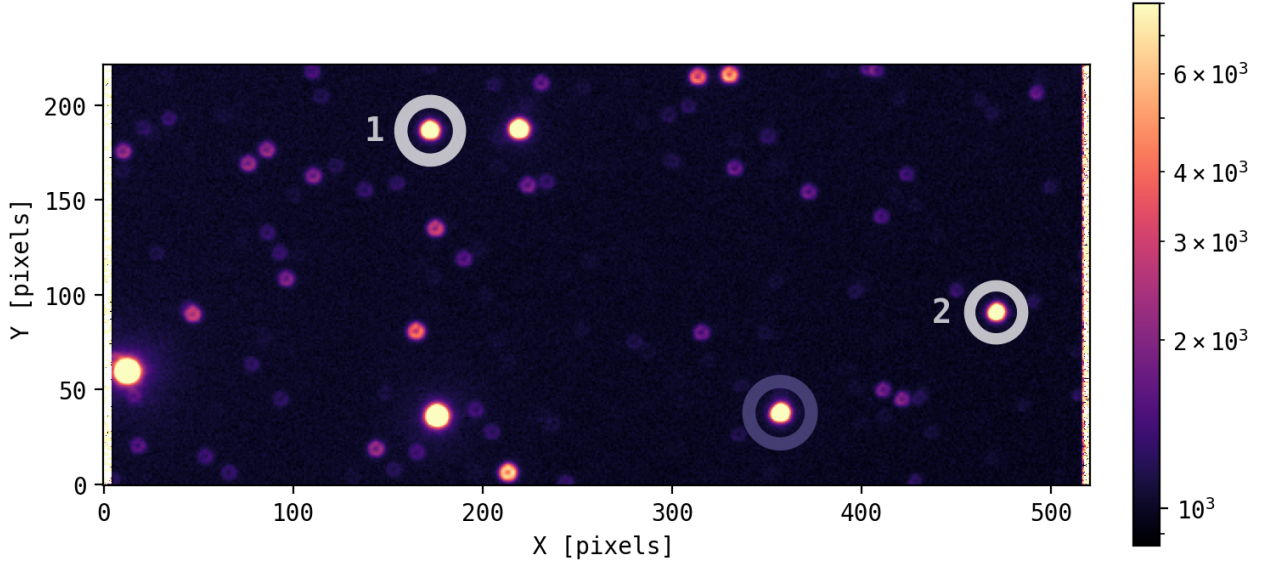


Fig. 1. Revised scientific framework marking the target star (purple annulus) with inner and outer radii of 13 and 20, respectively. The two reference stars, indicated by white annuli labelled 1 and 2, were used for differential aperture photometry. Reference 1 annulus has an inner radius of 12 and an outer radius of 19. Reference 2 annulus has an inner radius of 11 and an outer radius of 17. The colour bar represents the photoelectrons (intensity).

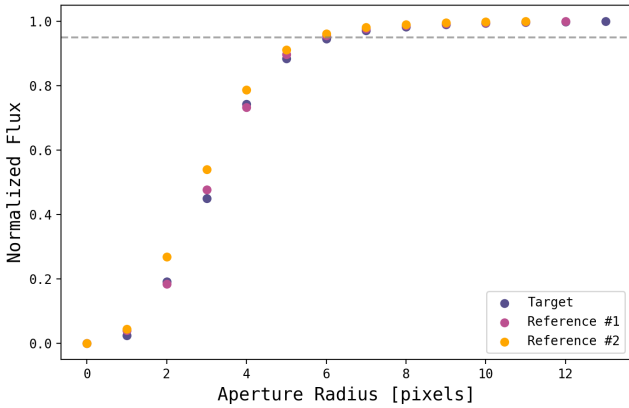


Fig. 2. A plot demonstrating Normalized Flux values against the Aperture Radii in pixel for the target and the two reference stars.

$$\sigma_{DF} = DF \sqrt{\left(\frac{\sigma_{\text{target}}}{F_{\text{target}}}\right)^2 + \sum \left(\frac{\sigma_{\text{reference}}}{F_{\text{reference}}}\right)^2}, \quad (15)$$

where σ_{target} and $\sigma_{\text{reference}}$ denote the uncertainties in the fluxes of the target and reference stars, respectively. The resulting differential light curve was examined for any outliers, which we identified and removed from subsequent analysis. In the final step, we normalized the light curve by dividing the differential flux by its median value, facilitating easier interpretation and comparison of transit events. This approach produced a high-precision light curve, well-suited for the modelling and analysis of transit signals.

3.2. TESS Dataset

3.2.1. Data Preprocessing

In this section, we examine the Light Curve files of our selected TESS sectors. TESS selects its aperture and background pixels from the available pixels in the Target Pixel file (TPF). It computes aperture photometry by summing the flux of multiple pixels within the designated aperture to yield a singular flux value for the star, while the background flux is subtracted by calculating the average background value per pixel using an alternative aperture. In the TESS dataset, the flux data with the term Simple Aperture Photometry (SAP) represent the data with aperture photometry that was done. While, the Pre-search Data Conditioning SAP (PDCSAP) flux is identical to the aperture photometry derived from the TPF, with the added elimination of long-term trends and instrumental systematics by the application of Co-trending Basis Vectors (CBVs). CBVs denote the collection of systematic trends evident in the ensemble flux data. CBVs are allocated for each operational area of the mission. The TESS pipeline obtains these from a Principal Component Analysis and used them to alleviate systematic artifacts in the target light curves. If TESS users observe lingering systematic issues in their Light Curve data, the CBVs can be utilised to conduct a manual photometric correction that is more specifically aligned with the user's scientific objectives. PDCSAP flux typically exhibits clearer data than SAP and demonstrates less systematic tendencies. Nonetheless, in certain instances, the correction may be ineffective; hence, we will examine both fluxes to evaluate the results of automatic corrections implemented by general-purpose pipelines.

During the investigation of the LCfs, we discovered that an offset has been eliminated from each timestamp. This

standard is commonly known as *Barycentric TESS Julian Date*. To accurately obtain the time array, it is necessary to convert BTJD to BJD by reintroducing the removed constant. We then plotted SAP and PDCSAP, revealing that the quantity of points in PDCSAP is less than that in SAP. Despite the identical configuration of the arrays for both fluxes and time. We acknowledge that the PDCSAP flux may contain NaN and infinite values; hence, we performed a "finite selection" on the data utilising `np.isfinite`. The dataset also contains Cadence Quality Flags recorded as a bitmask, which we use in our secondary selection to eliminate all points with any flag designated as True. This is accomplished by excluding observations with a quality marker exceeding zero. This method is referred to as pessimistic selection. Another selection is optimistic selection, in which we decide what errors prevent our observations from being considered. In the bitmask, flags intended for the exclusion of an observation are denoted as 1 in binary notation, whereas flags to be disregarded are represented as 0. In our instance, the pessimistic approach is favoured, as incorporating potentially problematic observations necessitates a comprehensive understanding of the instrument and the factors that result in the rejection of certain observations. We have also incorporated a time selection for if a transit happens at the beginning of the time series, with pre-transits entirely omitted due to the quality flag.

TESS data retains both instrumental and astrophysical traces in SAP data, while PDCSAP data has solely astrophysical signatures. Our methodology for eliminating these characteristics, when several weeks of continuous observations are accessible, involves the empirical removal of unwanted signals, irrespective of their sources, utilising an automatic filtering algorithm. This method is already incorporated in a package named `wotan`, which offers a user-friendly interface to evaluate several filtering algorithms. We filtered both SAP and PDCSAP light curves and compared the outcomes of the various techniques. We obtained the orbital period, central transit time, and transit duration (converted to days) of our planet from TESS ExoFOP. We conducted our initial test utilising the Huber spline within the flatten function, which serves to de-trend light curves by eliminating long-term variability stemming from stellar activity, instrumental noise, or systematic errors, thereby preserving only the short-term fluctuations such as those induced by transits or eclipses. We subsequently folded the transit around the centre time of transits, recognising that the identical signal is anticipated to recur after each orbital period. The plot illustrated that the segment of the light curve immediately preceding and following the transit is not centred around unity as expected. The reason is that the filtering method continues to incorporate certain points associated with the transit while calculating the model. Consequently, we calculated the filtering model by removing the points influenced by the transit, utilising a selection (mask) with a duration twice that of the transit window. This selection is solely utilised for the filtering phase. We subsequently tested the efficacy of the flattened masked flux against the non-masked variant by calculating the standard deviation of each flux, which led us to proceed with the masked flattened flux. We used two different algorithms (biweight and Huber spline), utilising two window lengths for each approach, and computed the standard deviation for all cases as presented in table .1. The selection of the flux was informed by an analysis of the minimal

standard deviation linked to specific algorithms and window sizes, alongside an evaluation of its overall quality as assessed through visualisation techniques. In Sector 43, we opted to use the masked PDCSAP flux processed with a Huber spline and a window size of 1.0, as this configuration provided the most favourable fit. For Sectors 44 and 45, the masked SAP flux was selected with a window size of 1.0. Although PDCSAP frequently demonstrated superior standard deviation values, our examination of the model fit and the corresponding STD values indicated that SAP offered a more precise representation for the last two sectors.

3.2.2. Transit Model Initialization

We are now generating a model of planetary transit within a light curve utilising parameters derived from existing research. We utilised the optimal TASTE and TESS light curves using the selection criteria previously established. We used the `batman` (Bad-Ass Transit Model cAlculation) for efficient and rapid computation of transit light curves. The published research ((Ozturk & Erdem 2019)) was utilised as a reference for the parameters required to model the transit:

- Time of inferior conjunction (T_0): 2454508.97824
- Orbital period (P): 1.0914199 days
- Planet-to-star radius ratio (R_p/R_*): 0.119
- Semi-major axis (a/R_*): 2.933
- Inclination (i): 81.92°

We presumed a circular orbit for the planet, establishing the orbital eccentricity at zero and determining the argument of periastron at 90 degrees. We computed quadratic limb-darkening coefficients (u_1 and u_2) utilising the Limb Darkening Toolkit (LDTk) to include stellar limb-darkening effects into our transit model. This calculation depended on stellar properties pertinent to our target star, encompassing effective temperature, surface gravity, and metallicity. In this investigation, we evaluated the coefficients for the TESS and TASTE (Sloan r) filters by creating limb-darkening profiles by Monte Carlo sampling with 10,000 iterations. The profiles were resampled with 100 linear increments throughout the stellar disc to guarantee adequate resolution. To mitigate any inadequacies in the theoretical models, we utilised an uncertainty multiplier of 10 when adjusting the derived coefficients. For the TESS passband, we determined $u_1 = 0.41 \pm 0.01$ and $u_2 = 0.13 \pm 0.04$. For the Sloan r filter, we determined $u_1 = 0.51 \pm 0.01$ and $u_2 = 0.13 \pm 0.03$. These characteristics were then included in our transit model. A star's surface has different temperatures and opacities; limb darkening makes the edges seem darker than the middle. Cooler, shallower layers provide light in the outer portions of the star, while deeper, hotter layers provide the brilliance in the centre. Photons absorb and scatter more as they come into contact with more star material near the edge, which reduces their intensity.

To reduce this effect, we utilised a quadratic limb-darkening equation, which harmonises accuracy and simplicity in predicting star intensity fluctuations. The following equation:

$$I(\mu) = I_0 (1 - u_1(1 - \mu) - u_2(1 - \mu)^2), \quad (16)$$

where $I(\mu)$ denotes the specific intensity at a particular angle, I_0 signifies the intensity at the centre of the stellar disc, and $\mu = \cos(\theta)$, with θ representing the

angle between the observer's line of sight and the normal to the stellar surface. The coefficients u_1 and u_2 were obtained from the stellar parameters of WASP-12 utilising the limb-darkening tables supplied by Claret (2004). The integration of these factors into the transit model enhanced the precision of the light curve fit and guaranteed that the estimated planetary parameters, such as the radius ratio (R_p/R_*), were not skewed by unconsidered stellar intensity fluctuations.

4. Analysis

In this section, we describe the approach used to model and assess the transit light curves from the TESS and TASTE datasets. This involved creating a transit model, including prior understanding of key features, optimising these parameters using Bayesian inference, and confirming the model using statistical analysis and visualisation. We utilised the `batman` library to model the transit light curve and calculate theoretical light curves. We created distinct transit models for each sector of the TESS dataset utilising an identical parameter set. The light curves from each area were computed and subsequently amalgamated to form a cohesive dataset for examination. This method enabled us to consider sector-specific differences while utilising the complete observational data. We combined the time and the flux data from all sectors into a singular dataset for model fitting. For the TASTE dataset, we integrated a second-order polynomial trend to address systematic effects. The trend was parametrized in relation to the median observation time, assuring model consistency and stability. The integrated model for TASTE encompassed the polynomial trend and the theoretical transit model.

We assessed the fit quality between our model and the data by calculating the log-likelihood of the model's parameters in relation to the data. The log-likelihood function integrates the chi-squared statistic, the entire count of data points, and the error terms associated with both the TESS and TASTE datasets:

$$\ln L = -0.5 \left[N \ln(2\pi) + \chi^2 + \sum \ln(\sigma^2) \right] \quad (17)$$

Where N denotes the total number of data points, χ^2 signifies the sum of squared deviations between the observed and modelled flux values (normalised by their errors), and σ encompasses both measurement uncertainties and jitter parameters.

We implemented the log-likelihood calculation to include data from all TESS sectors as well as the TASTE dataset. We determined the total chi-squared value (χ_{total}^2) by adding up the chi-squared values from each individual dataset. The log-likelihood value we obtained, $\ln L = 73661.3834$, indicates the degree of consistency between the model and the observed data. Calculating the log-likelihood is important as it enables a quantitative evaluation of the extent to which the model parameters accurately represent the observed data. The log-likelihood serves as a foundational element for calculating posterior probabilities when integrated with prior distributions in Bayesian inference.

Priors were applied to specific parameters, which are limb darkening coefficients, which were modelled as Gaussian distributions informed by theoretical predictions. The

TESS limb darkening coefficients u_1 and u_2 , and the TASTE coefficients u_1 and u_2 , all exhibit a standard deviation of 0.10. The priors guaranteed that the exploration of parameters stayed within the bounds of physical significance and adhered to recognised astrophysical constraints. We established boundaries on the parameter space to prevent unphysical combinations of parameters. For example, the parameters including the time of inferior conjunction (T_0) and the orbital period (P) were limited to specific ranges based on the observational data. The coefficients for limb darkening were constrained within the range of 0.00 to 1.0, while the parameters for jitter were limited to reasonable values, such as 0.00 to 0.3 for TESS jitter. The boundaries effectively prevented the sampler from investigating areas of parameter space that could lead to the generation of invalid models.

The posterior probability was computed by the equation: $\ln P(\theta|D) = \ln P(D|\theta) + \ln P(\theta)$. In this context, $\ln P(D|\theta)$ signifies the log-likelihood, whereas $\ln P(\theta)$ indicates the sum of the log-priors. The restrictions were crucial in demonstrating that $\ln P(\theta|D) = -\infty$ for any parameter configuration that violated the limits, thereby systematically eliminating such samples.

The integration of priors and boundaries enabled a targeted exploration of parameters within regions that are both physically relevant and supported by observational data.

We initiated the estimation of the posterior distributions of the parameters by optimising the parameter space through the Differential Evolution optimiser available in the `pyde` package. The optimiser systematically examined the parameter space within the established limits to identify a collection of solutions that enhanced the posterior probability. The optimiser was initialised using 28 walkers along with the established parameter boundaries. We executed the optimiser for 1,000 generations to guarantee adequate exploration and convergence. The final population obtained from the `pyde` optimisation functioned as the first reference for the MCMC sampling procedure. We designed the sampler with 28 walkers and 20,000 steps utilising `emcee`, facilitating extensive sampling of the posterior distributions. By initialising the walkers with optimum beginning sites, we reduced the burn-in phase and improved the efficiency of the sampling procedure. The convergence of the MCMC chains was evaluated by analysing their traces. Subsequent to the elimination of the burn-in phase and the application of thinning, we standardised the chains and obtained a set of posterior samples. The posterior distributions of the model parameters were depicted using a corner plot (fig.1), clearly demonstrating the marginal distributions and the relationships between the parameters. The marginal distributions along the diagonal suggest that the majority of parameters are effectively constrained and display characteristics akin to Gaussian distributions. This indicates an effective alignment of the sampling process with reliable parameter estimation. Some of the off-diagonal plots indicate significant correlations among specific parameters.

There are moderate correlations between the limb-darkening coefficients u_1 and u_2 in both the TESS and TASTE datasets, indicating their interconnected influence on the transit light curve. Polynomial coefficients (c_0 , c_1 , c_2) for the TASTE dataset exhibit minimal correlation, indicating that the polynomial model effectively captures systematic trends. The narrow distributions observed for

key parameters like T_c (T_0) and P highlight the robustness of the integrated dataset and model. The corner plot offers a detailed examination of the posterior space, highlighting the model's effectiveness in accurately representing the dynamics of the transit light curve.

The reconstructed models were validated through a comparative analysis with the observed flux data obtained from the TESS and TASTE datasets. The flux generated by the TASTE model, which includes the polynomial trend, was layered over the actual TASTE data observed. The scatter plot .2 that compares the observed flux with the modelled flux shows a strong correlation, confirming the polynomial trend's effectiveness in addressing systematic variations. The integrated model for all three TESS sectors was layered onto the consolidated dataset, revealing a significant alignment with the observed data. The TESS light curve was phase-folded around the orbital period to enhance the visibility of the transit features. The time data underwent folding and shifting to align the transit event at the centre. In order to consolidate the out-of-transit data and highlight the key transit characteristics, this step methodically aligned all transit observations. The phase-folded model, as illustrated in fig .3, was overlaid on the data, facilitating a direct comparison between the theoretical model and the observed results. The phase-folded TESS light curve indicates a subtle misalignment between the observed data and the modelled transit, suggesting that the model is marginally shifted to the right. This discrepancy probably arises from an offset in the estimated time of inferior conjunction (T_0) or from residual systematics present in the combined TESS dataset which were visualized in fig .3. The overall alignment lines up with the expected transit features; however, this phase shift underscores the necessity for further refinement in estimating T_0 or for additional de-trending of the TESS data to reduce residual effects.

We calculated essential statistical metrics to summarise the posterior distributions of the parameters, consisting of the mean, standard deviation, median, and 68% credible intervals ($\pm 1\sigma$ errors). The statistics offer a detailed examination of the parameter estimates derived from the MCMC sampling process. Regarding each parameter: The mean signifies the central tendency of the posterior samples, providing an average value for analysis. The Standard Deviation (Std) quantifies the dispersion of the posterior distribution, serving as an indicator of uncertainty. The median serves as the central value of the posterior distribution. $\pm 1\sigma$ denote the boundaries of the 68% credible interval, which is calculated by subtracting the median from the 16th and 84th percentiles. The results are presented in Table .2.

We calculated parameters, such as the impact parameter (b), planet radius (R_p), transit depth, transit signal-to-noise ratio (SNR) and, planetary density (ρ_p) by deriving them from posterior samples alongside supplementary stellar properties. The impact parameter, denoting the projected distance between the planet and the stellar centre expressed in terms of the stellar radius, was calculated using this equation ($b = \frac{a}{R_*} \cos(i)$) where all the parameters are known and i is expressed in radians. The analysis of the posterior samples for a/R_* and i yielded a mean and standard deviation for b of 0.408 ± 0.036 . The geometry of this

near-central transit configuration guarantees a robust transit signal. The planet radius (R_p) was calculated by utilising the ratio R_p/R_* obtained from the posterior samples, alongside the stellar radius ($R_* = 1.65 \pm 0.001 R_\odot$). We calculated the radius of the planet, in Jupiter radii (R_J), through ($R_p = \left(\frac{R_p}{R_*}\right) \cdot R_*$). Our measured planetary radius was $1.88 \pm 0.016 R_J$, which aligns with the characteristics of a gas giant. The transit depth is characterised by the square of the radius ratio (R_p/R_*), which effectively measures the proportion of stellar light obstructed by the planet throughout the transit event. The analysis of the posterior samples yielded a mean transit depth of 0.013715 ± 0.000152 , indicating a distinctly observable event in the light curve. The transit signal-to-noise ratio (SNR) was calculated by taking the mean transit depth and dividing it by the root mean square (RMS) of the residuals that exist between the observed and modelled TESS fluxes. The calculated SNR of 7.70 indicates a robust detection of the transit, affirming the dependability of the observed data and the model fit. The planetary density (ρ_p) The calculated planetary density was (), aligning with the anticipated values for the host star. The derived parameters offer additional validation of the transit model, affirming the planetary characteristics of the object, which is identified as a gas giant transiting a star in a near-central orbit. The elevated signal-to-noise ratio and precisely defined characteristics of both stellar and planetary bodies emphasise the strength of the analysis. The planetary density was obtained $\rho_p = 0.263 \pm 0.069 \text{ g/cm}^3$.

5. Discussion

TESS mission and TASTE project datasets were used to analyse WASP-12b transit light curves. Complementary observations increased temporal coverage, parameter estimations, and validation against independent datasets. Our findings emphasise the significance of integrating space-based and ground-based data for transit analysis, as emphasised by (Ozturk & Erdem 2019). We used TESS and TASTE datasets to refine planet-to-star radius ratio (R_p/R_*) and orbital period (P) while preserving agreement with previous research. Our results were reliable and robust thanks to our multi-instrument method. TESS data from three sectors recorded transit characteristics with excellent precision and continuous observations. Multiple sectors were needed to account for temporal changes and provide a complete transit picture. Additional ground-based observations adjusted for systematic trends using a second-order polynomial were provided by the TASTE dataset. These datasets provided a solid foundation for transit light curve modelling and planetary parameter derivation. We used Bayesian inference to improve model parameters utilising theoretical priors and data. The use of `pyde` for initial optimisation and `emcee` for posterior sampling facilitated a comprehensive examination of the parameter space. The corner plot demonstrates a concordance between the model and the observed data, evidenced by a high log-likelihood value ($\ln L = 73661.3834$) and well-defined posterior distributions. Nevertheless, the phase-folded TESS light curve showed a minor discrepancy between observed data and model. The observed discrepancy may arise from residual systematics in the TESS data, including spacecraft jitter or thermal effects, as indicated by Van Cleve (2016). It

may also be associated with orbital decay or perturbations caused by additional companions, as indicated in other systems by Maciejewski (2018). Resolving this issue necessitates the inclusion of supplementary TESS sectors or the integration of transit timing variations (TTVs) with radial velocity measurements to enhance T_c estimates and examine long-term orbital evolution. We determined the planetary radius to be $1.88 \pm 0.016 R_J$, a transit depth of 0.013715 ± 0.000152 , and an impact parameter of 0.408 ± 0.036 . The findings perfectly align with Ozturk & Erdem (2019), very few discrepancies were found (table .3). Our study found that the semi-major axis ratio (a/R_s) and orbital inclination (i) are close to the literature estimates, with $a/R_* = 2.979940 \pm 0.041672$ vs 2.933 and $i = 82.145330 \pm 0.688439$ versus 81.92 . These small variances may be due to modelling assumptions, parameter initialisation, or data processing. Our results are robust, although the discrepancies need refining. Adding more TESS sectors or independent datasets might improve parameter accuracy and resolve discrepancies. Such investigations might illuminate WASP-12b's orbital dynamics. One notable aspect of our analysis is the derived planetary density (ρ_p) of WASP-12b, which is lower than the value reported in (Ozturk & Erdem 2019). This discrepancy may arise from systematic differences in light curve modelling, variations in limb-darkening coefficients, or the inclusion of additional observational datasets in our analysis. However, beyond methodological factors, this result may also reflect the ongoing physical evolution of the planet.

WASP-12b is known to exhibit an inflated planetary structure, which aligns with previous studies suggesting that it is undergoing atmospheric escape due to extreme stellar irradiation (Li et al. 2010). Several physical mechanisms contribute to planetary inflation, including intense stellar irradiation, Ohmic heating, tidal heating, and slow interior cooling. Due to its extreme proximity to its host star, WASP-12b is highly irradiated and tidally heated, making it one of the most inflated hot Jupiters observed to date.

Additionally, WASP-12b is believed to be experiencing Roche-lobe overflow (RLOF), a process in which mass is stripped from a planet's outer atmosphere when it exceeds its Roche lobe, the region where the planet's gravity can still hold onto its gas. Because of its large, extended atmosphere, WASP-12b is likely at or near the point of overflowing this critical boundary. Observations suggest that the planet is actively losing mass over time, meaning that it is slowly being accreted by its host star. This mass loss process not only alters the planet's atmospheric composition but can also reduce its overall density.

If WASP-12b continues to lose mass at the rate predicted by (Li et al. 2010), its density should gradually decrease over time. Thus, the discrepancy between our density estimate and previous literature values could reflect this ongoing mass loss, rather than just differences in modelling approaches. The planet's physical properties may have evolved, contributing to the observed lower density. Future work incorporating high-resolution transmission spectroscopy and additional photometric datasets could further constrain the extent of mass loss and refine estimates of WASP-12b's atmospheric escape rate, providing deeper insight into its ongoing physical transformation.

6. Conclusions

This study entailed conducting aperture photometry on the TASTE data, implementing corrections for bias and flat-field effects, and concluding with differential photometry. In addition to implementing filtering on the TESS sector's data to enhance the quality of the light curves. The filtering included removing values like infinite and undefined points, points with quality flag higher than zero and transits after previous data was filtered out. The model needed this preprocessing step to ensure dataset integrity and properly collect transit characteristics.

We used Bayesian inference and Markov Chain Monte Carlo (MCMC) sampling to obtain well-constrained posterior distributions for essential planetary parameters. The planetary radius is determined to be $1.88 \pm 0.016 R_J$, with a transit depth of 0.013715 ± 0.000152 , aligning with the properties of a gas giant. The orbital period has been refined to $P = 1.091418 \pm 0.000001$ days, providing enhanced precision relative to previous studies.

Despite little variations in orbital inclination (i), our findings are largely consistent with earlier studies, including Ozturk & Erdem (2019). Different modelling assumptions, parameter initialisation, or dataset preparation may cause variations. Phase-folded TESS light curves showed a minor mismatch between observed data and model, perhaps due to residual systematics or an offset in inferior conjunction (T_0). These results emphasise the necessity of enhancing parameter estimation and detrending.

The difference between the density obtained in our research and value in the literature could reflect the physical evolution of the planet, where ongoing mass loss has decreased the planet's density over the past years. Future work incorporating transmission spectroscopy and additional photometric datasets could provide further insight into these atmospheric processes.

Finally, Our research lays a strong basis for WASP-12b and related planetary system investigations and highlights the value of multi-instrument approaches in exoplanet research. Aperture photometry and light curve filtering are crucial data processing techniques for precise and trustworthy research.

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Table .1. Standard Deviations (STD) of SAP and PDCSAP fluxes for Sectors 43, 44, and 45 using different algorithms and window sizes.

Sector	Flux Type	Algorithm	Window Size	STD
43	SAP	Biweight	1.0	0.001917
43	SAP	Biweight	1.5	0.001919
43	SAP	Hspline	1.0	0.001917
43	SAP	Hspline	1.5	0.001920
43	PDCSAP	Biweight	1.0	0.001884
43	PDCSAP	Biweight	1.5	0.001886
43	PDCSAP	Hspline	1.0	0.001884
43	PDCSAP	Hspline	1.5	0.001884
44	SAP	Biweight	1.0	0.001882
44	SAP	Biweight	1.5	0.001884
44	SAP	Hspline	1.0	0.001881
44	SAP	Hspline	1.5	0.001883
44	PDCSAP	Biweight	1.0	0.001893
44	PDCSAP	Biweight	1.5	0.001896
44	PDCSAP	Hspline	1.0	0.001893
44	PDCSAP	Hspline	1.5	0.001896
45	SAP	Biweight	1.0	0.001835
45	SAP	Biweight	1.5	0.001837
45	SAP	Hspline	1.0	0.001836
45	SAP	Hspline	1.5	0.001837
45	PDCSAP	Biweight	1.0	0.001848
45	PDCSAP	Biweight	1.5	0.001851
45	PDCSAP	Hspline	1.0	0.001848
45	PDCSAP	Hspline	1.5	0.001854

Table .2. Posterior Parameter Statistics from MCMC Analysis

Parameter	Mean	Std Dev	Median	+1 σ Error	-1 σ Error
T_c (Mid-Transit Time)	2.454509×10^6	0.006094	2.454509×10^6	0.001430	0.001721
P (Orbital Period)	1.091418	0.000001	1.091418	0.0000004	0.0000003
R_p/R_* (Planet-Star Radius Ratio)	0.117112	0.000649	0.117206	0.000420	0.000533
a/R_* (Semi-Major Axis Ratio)	2.979940	0.041672	2.984602	0.029869	0.032248
i (Orbital Inclination)	82.145330	0.688439	82.163250	0.592891	0.568886
$u_{1,\text{TESS}}$ (Limb Darkening Coeff. 1)	0.286128	0.050580	0.290297	0.040813	0.042119
$u_{2,\text{TESS}}$ (Limb Darkening Coeff. 2)	0.189133	0.103942	0.174719	0.079193	0.070838
$u_{1,\text{TASTE}}$ (Limb Darkening Coeff. 1)	0.433667	0.050335	0.435043	0.044274	0.049273
$u_{2,\text{TASTE}}$ (Limb Darkening Coeff. 2)	0.1219312	0.086568	0.106114	0.084245	0.062655
c_0 (Polynomial Coefficient 0)	0.761335	0.000290	0.761401	0.000072	0.000086
c_1 (Polynomial Coefficient 1)	0.002105	0.002335	0.002648	0.000810	0.000997
c_2 (Polynomial Coefficient 2)	0.053286	0.074908	0.035666	0.019708	0.017053
J_{TESS} (TESS Jitter)	0.001862	0.000008	0.001862	0.000007	0.000007
J_{TASTE} (TASTE Jitter)	0.001678	0.000086	0.001658	0.000042	0.000032

Table .3. Comparison of derived parameters for WASP-12b between this work and the literature (Ozturk & Erdem 2019).

Parameter	This Work	Uncertainty	Literature Value	Difference
T_c (Mid-Transit Time) [BJD]	2454509.000	± 0.006094	2454508.97824	+0.02176
P (Orbital Period) [days]	1.091418	± 0.000001	1.0914199	-0.0000019
R_p/R_* (Planet-Star Radius Ratio)	0.117112	± 0.000649	0.119	-0.001888
a/R_* (Semi-Major Axis Ratio)	2.979940	± 0.041672	2.933	+0.046940
i (Orbital Inclination) [degrees]	82.145330	± 0.688439	81.92	+0.225330
ρ_p (Planetary density) [g/cm ³]	0.263	± 0.069	0.311	-0.048

Notes: The literature values are sourced from Ozturk & Erdem (2019). T_c and T_0 are the same. Differences are computed as "This Work - Literature Value". The inconsistency in i indicates a possible modelling error, as transiting planets generally possess inclinations $< 90^\circ$.

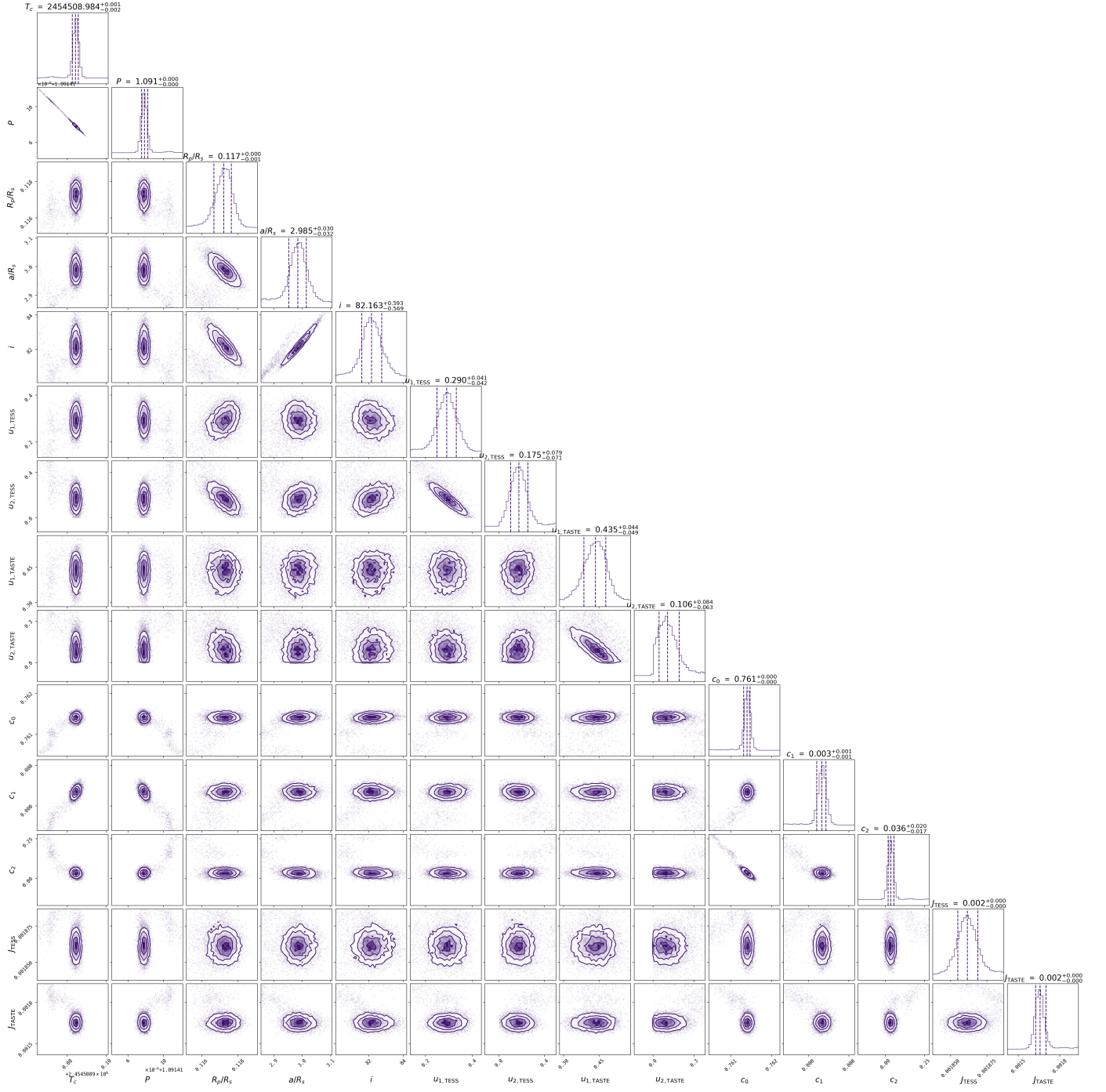


Fig. .1. Corner plot illustrating the posterior distributions and the correlations between parameters. The diagonal panels illustrate the marginalised posterior distributions, whereas the off-diagonal panels present the pairwise correlations.

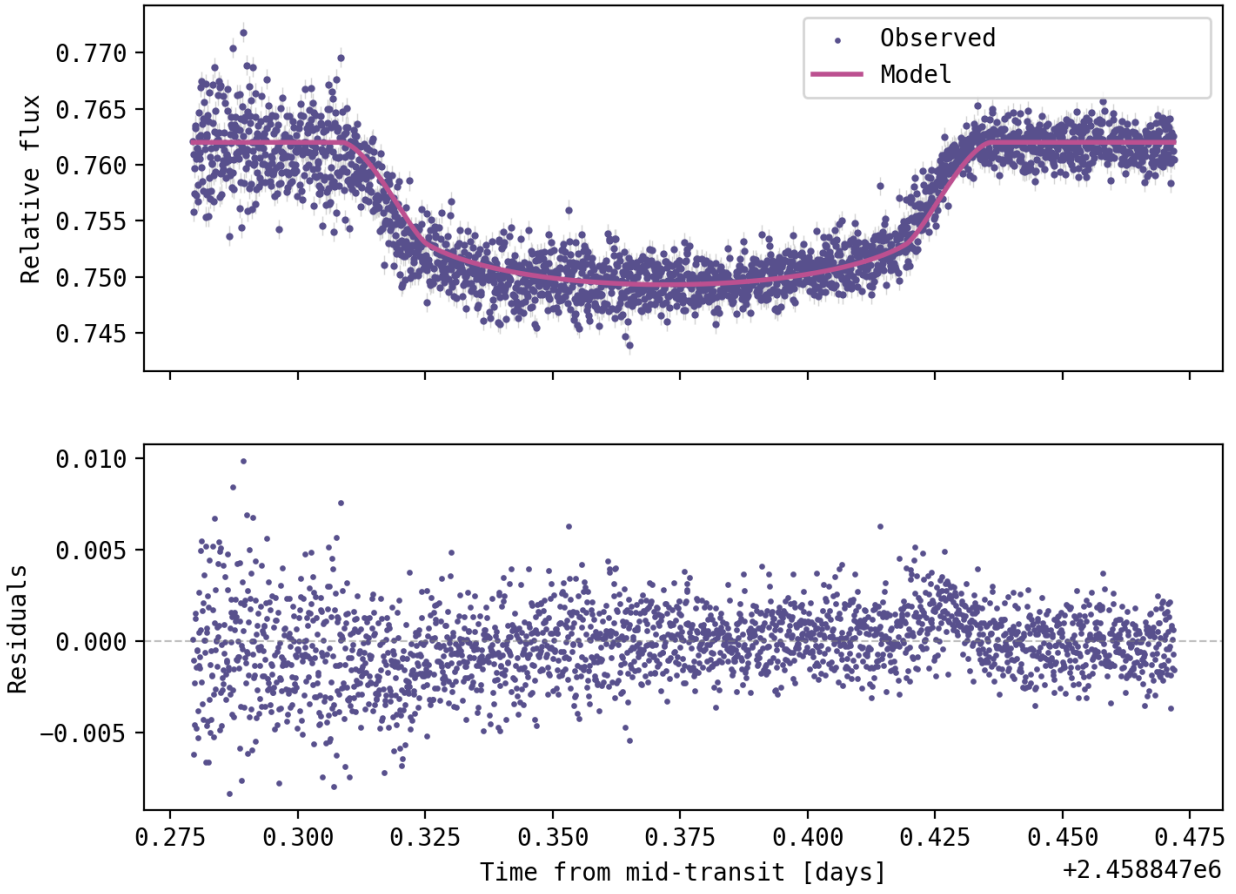


Fig. .2. Top: The observed TASTE flux (points) is compared with the best-fit transit model (solid line) with errors included. Bottom: The residuals represent the difference between the observed flux and the model flux, illustrating the strong correlation between the data and the model, characterised by minimal scatter.

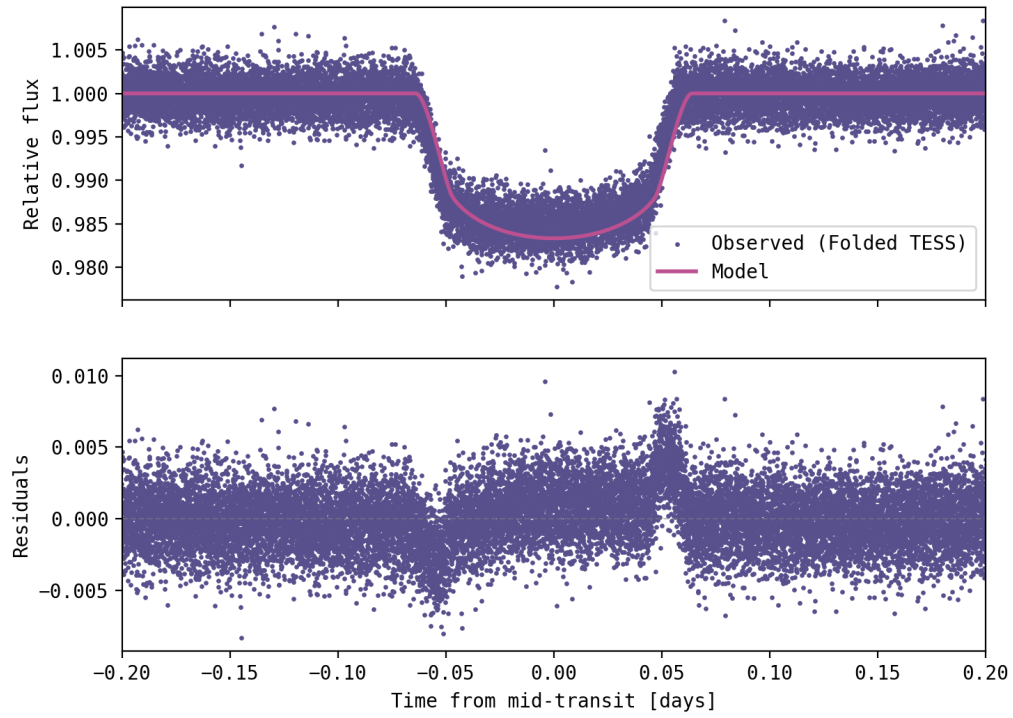


Fig. .3. Top: The observed time-folded TESS flux (points) compared with the best-fit transit model (solid line) within combined sectors. Bottom: The residuals represent the difference between the observed flux and the model flux, showing minimal scatter with some systematic structure near the transit edges.